

GMaster: a difference-form Wigner- d march for GPU MASTER estimation

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ABSTRACT

Masked full-sky maps are analysed through spherical harmonic transforms and pseudo- C_ℓ (MASTER) estimators that propagate survey windows into mode coupling. On GPUs the latitudinal factor of a HEALPix transform is often evaluated as a serial Jacobi recurrence in degree; in float32 that recurrence is unstable near the poles and at turning points, where a rounding of size $u|v|$ is amplified by $1/\sin\phi$. We rewrite the march so that the carried pair is (v_n, D_n) with $D_n = v_n - v_{n-1}$ rather than (v_n, v_{n-1}) . The difference coefficient vanishes as ϕ^2 exactly where the plain form is ill-conditioned, error is injected at the scale of $D \sim \phi v$, and the float32 floor becomes the random walk $\sqrt{n}u$. Against the float64 recurrence at $N_{\text{side}} = 1024$ the maximum relative error over every lane falls from 4.2×10^{-2} (plain) to 2.8×10^{-6} (shipped). The identities of the rewrite, a southern-hemisphere reflection onto the same code path, and an exact per-block power-of-two rescale are proved in Lean 4. The kernel is CUDA, called from JAX through the XLA FFI, and is the default latitudinal operator in GMaster, a GPU library with the public API of NaMaster. We benchmark GMaster against widely used state-of-the-art methods and find 20 $\times$ improvement in end-to-end spin-0 MASTER estimation at $N_{\text{side}} = 1024$, 22 $\times$ at 2048, and 12 $\times$ at 8192, relative to NaMaster 3.0 on 96 CPU cores, at an accuracy loss below 10^{-3} for $\ell < 2N_{\text{side}}$ and up to 7×10^{-3} in the highest bandpowers at $N_{\text{side}} = 8192$, well below CMB map noise levels.

Key words: methods: numerical – cosmology: cosmic microwave background – cosmology: large-scale structure of Universe

1 INTRODUCTION

The MASTER estimator (Hivon et al. 2002) recovers binned angular power spectra from masked spherical maps by analysing the masked fields, forming coupled pseudo-spectra, building a mode-coupling matrix from the mask, and decoupling. Throughout, ℓ and m denote the usual spherical-harmonic multipole and azimuthal order, and $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ are colatitude and azimuth on the sphere.

NaMaster (Alonso et al. 2019) is the standard CPU implementation of that pipeline: `pymaster` 3.0 exposes `NmtField`, `NmtWorkspace`, and decoupled bandpowers with the normalisations used in current CMB and LSS analyses. Its HEALPix transforms call DUCC0 (Reinecke & Seljebotn 2013), the successor of libsharp, on the HEALPix pixelisation (Górski et al. 2005). The coupling algebra has a structured Wigner- $3j$ recurrence (Kiddier & Gratton 2026). We time NaMaster 3.0 with `ducc0` 0.39.1 on the same host as GMaster.

SHTns (Schaeffer 2013) is a widely used pseudospectral library for on-the-fly associated-Legendre recurrences; it now includes a CUDA implementation (<https://nschaeff.bitbucket.io/shtns/>). Our transform timings use that GPU module (`SHT_ALLOW_GPU`) from a current source build wired

in `benchmarks/scaling/bench_three_way.py`, not the 2013 journal release in isolation. SHTns has no spin-2 transform; spin-2 panels time its spin-1 vector transform as a GPU reference. GMaster is not benchmarked as a drop-in replacement for arbitrary SHTns grids; the comparison is isolated `alm2map/map2alm` passes at matched $\ell_{\text{max}} = 3N_{\text{side}} - 1$.

On a GPU the same two cubic stages remain. An on-the-fly recurrence, as in SHTns and DUCC, is a serial chain in ℓ with a reduction over colatitude at every degree. That pattern matches AVX-512 CPUs and mismatches GPU SIMT. A precomputed Legendre matrix, as in the precompute mode of `s2fft` (Price & McEwen 2024), inverts the bottleneck provided the table fits. Related GPU work (Belkner et al. 2024; Reinecke et al. 2023) retains an on-the-fly latitudinal factor and is not a drop-in for a full-sky HEALPix MASTER pipeline.

GMaster is a JAX implementation of the NaMaster public API. This Letter isolates one change that made a table-free float32 latitudinal kernel accurate enough to be the default at every $N_{\text{side}} \geq 64$: marching the difference pair of the Jacobi row attached to $d_{m,-s}^\ell(\theta)$ rather than the consecutive values. The recurrence, a reflection that sends the southern hemisphere through the same instructions, and the exactness of a per-block power-of-two rescale are proved in Lean 4 (de Moura & Ulrich 2021; The mathlib Community 2020). We report the end-to-end MASTER board and two real-map

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overlays. This is not a cosmological analysis: no beam, split cross-spectrum, or covariance is applied to the survey maps.

2 DIFFERENCE-FORM MARCH

Every Wigner function $d_{m,-s}^\ell(\theta)$ is an algebraic multiple of a Jacobi polynomial (Olver et al. 2010, NIST DLMF 14.9.16, 18.3),

$$d_{m,-s}^\ell(\theta) = (-1)^m 2^{\varepsilon_m \text{LG2N}} \times \left(\sin \frac{\theta}{2}\right)^\alpha \left(\cos \frac{\theta}{2}\right)^\beta P_n^{(\alpha,\beta)}(\cos \theta), \quad (1)$$

with $\alpha = m + s$, $\beta = |m - s|$, $n = \ell - \max(m, s)$ and $\text{LG2N} = \log_2 \sqrt{(\ell + m)! (\ell - m)! / ((\ell - s)! (\ell + s)!)}$. The family obeys a three-term recurrence in n (Olver et al. 2010, DLMF 18.9.1). Writing $x = \cos \theta$ and v_n for the marched polynomial,

$$v_n = A_n v_{n-1} - B_n v_{n-2}, \quad A_n = c_1 x + c_0, \quad B_n = c_2, \quad (2)$$

where c_1, c_0, c_2 depend only on (ℓ, m) . A float64 march of (2) agrees with a numpy oracle of the same coefficients to 1.5×10^{-15} at $N_{\text{side}} = 1024$. Plain float32 of the same recurrence reaches a maximum relative error of 4.2×10^{-2} over the row (Table 1).

Over a few degrees the coefficients are nearly constant and $c_2 \rightarrow 1$, so the row is locally $v_n \approx A v_{n-1} - v_{n-2}$ with characteristic roots $e^{\pm i\phi}$ and $2\sqrt{c_2} \cos \phi = A$. In the oscillatory region $|A| < 2$, and $\phi \rightarrow 0$ both at the poles and at every turning point $m \approx (\ell + \frac{1}{2}) \sin \theta$. A rounding $\delta \sim 2u|v|$ of $\text{fl}(A v_{k-1} - B v_{k-2})$ perturbs the state by $(\delta, 0)$, which decomposes on the modes $(e^{\pm i\phi}, 1)$ with amplitude $\delta/(2 \sin \phi)$ each and evolves as $\delta \sin((n - k + 1)\phi)/\sin \phi$: every rounding is amplified by $1/\sin \phi$, and over n steps with random signs the relative error is $\sim \sqrt{n} u / \sin \phi$. On the first HEALPix ring at $N_{\text{side}} = 4096$, $\theta \approx 2 \times 10^{-4}$ and $1/\sin \phi \approx 5 \times 10^3$, giving a floor of a few per cent, the order of the plain-float32 row of Table 1. A systematic error is worse: a relative error ϵ in x perturbs A at every step, and $2 \cos \phi = A$ turns it into a coherent phase drift $\sim n c_1 x \epsilon / (2 \sin \phi)$, of order a radian at $n = 12288$ for $\epsilon = u$.

Let $D_n = v_n - v_{n-1}$. Then for $n \geq 3$

$$D_n = (A_n - 1 - B_n) v_{n-1} + B_n D_{n-1}, \quad v_n = v_{n-1} + D_n. \quad (3)$$

Equation (3) is the central result of this Letter: the latitudinal march is carried in difference form, and every subsequent kernel design choice (two-word C , reflection, rescale) is in service of making that recurrence stable in float32. In terms of ϕ , the difference coefficient is $C = A - 1 - c_2 \approx -\phi^2$, so $D \approx \phi |v|$ exactly where the plain march is ill-conditioned. A rounding of $v \leftarrow \text{fl}(v + D)$ lands on v with D fixed and is not amplified. A rounding of $D \leftarrow \text{fl}(c_2 D + C v)$ lands on D at scale $u\phi|v|$ and, after the $1/(2 \sin \phi)$ modal factor, contributes $O(u|v|)$ per step. Both kinds therefore give a random-walk floor $\sqrt{n} u$ independent of ϕ : $\sqrt{3072} u \approx 3.3 \times 10^{-6}$ at $N_{\text{side}} = 1024$, against the measured 2.8×10^{-6} .

The only coefficient on the amplified path is C , and the argument needs its error to be $O(u|C|)$, not $O(u|A|)$. Writing $C_n = c_1(x - 1) + E_n$ with $E_n = c_1 - 1 - c_2 + c_0$, the kernel carries $x - 1$ rather than x : near a pole $x - 1 \sim -\theta^2/2$, so a single-word x would inject a relative error $u/(\theta^2/2)$ into the leading term of C , and a single-word E would leave an error

Table 1. Maximum relative error of a float32 march against the float64 recurrence, over every lane, at $N_{\text{side}} = 1024$ spin 0. The two-word row is the shipped kernel (5.7×10^{-6} at $N_{\text{side}} = 4096$).

Float32 march	$m = 0$	$m = 1000$	$m = L/2$
Plain, single word	4.2×10^{-2}	8.8×10^{-5}	3.1×10^{-5}
Difference form, 1-word C	4.1×10^{-5}	8.6×10^{-5}	5.1×10^{-5}
Difference form, 2-word C	2.8×10^{-6}	2.9×10^{-6}	1.6×10^{-5}

$\sim u|c_1|$ that does not shrink with C . Both are systematic and drift the phase. The kernel therefore stores $|x| - 1 = x_h + x_l$, $c_1 = c_{1h} + c_{1l}$ and $E = E_h + E_l$ as float32 pairs, each split from float64 to 2^{-48} (Dekker 1971), and forms

$$C = \text{fma}(c_{1h}, x_h, E_h) + [\text{fma}(c_{1h}, x_l, \text{fma}(c_{1l}, x_h, E_l))]. \quad (4)$$

The first **fma** rounds once, by $u|C|$, and that rounding vanishes with C at the turning point where the amplification peaks; the bracket is $\sim 2^{-24}$ of the leading terms. A single-word C leaves the coherent $u|c_1|$ -class drift, growing like n rather than $\sqrt{n}$: the 4×10^{-5} floor in the middle row of Table 1. The coefficient c_2 is single-word, because its error enters as $u|D| \sim u\phi|v|$ on D , the unamplified kind.

The southern hemisphere needs no second code path. If $v_n = (c_1 x + c_0) v_{n-1} - c_2 v_{n-2}$ then $w_n := (-1)^n v_n$ satisfies

$$w_n = (c_1(-x) - c_0) w_{n-1} - c_2 w_{n-2}, \quad (5)$$

the same recurrence with $y = |x|$ for x and $-c_0$ for c_0 , whose difference coefficient is $c_1(y - 1) + E_n^-$ with $E_n^- = c_1 - 1 - c_2 - c_0$. Every lane, northern or southern, therefore marches $|x| - 1 \in (-1, 0]$ with the same instructions and one of two table columns $E^\pm$, and the emit restores the sign on odd n . Each lane stores $v \cdot 2^{ex}$ with ex an integer and is rescaled by an exact power of two every eight degrees; the map $(v, D) \mapsto (v + D', D')$ is linear, so the represented value is unchanged, and the per-degree growth is bounded by $1 + |C| + c_2 \leq 2^{15}$, so eight degrees cannot overflow. The emit factorises the Jacobi normalisation $2^{\varepsilon_m \text{LG2N}}$ into a per-lane power of two and a per-degree mantissa proved to lie in $[2^{-45.5}, 2)$, so no exponent lane leaves the kernel. Equations (3) and (5), the seeds, the homogeneity of the rescale, and the range of the mantissa are theorems in a Lean 4 development on Mathlib that builds with no **sorry**; the modal error argument is not formalised, and is checked by Table 1.

The kernel is CUDA, one warp per $(m, \text{ring tile})$, exposed as XLA FFI handlers inside **jit**. A field and its mask are two transforms of the same rows: the analysis kernel takes both as emit channels, so an **NmtField** is not two launches. For $s = 0$ the southern hemisphere is folded into the northern recurrence. The default at $L = \ell_{\text{max}} + 1 \geq 192$ ($N_{\text{side}} \geq 64$) is this march; an exact float64 route remains available by flag. The difference-form D_n march is not GPU-intrinsic; only the shipped latitudinal operator is CUDA. A fair CPU timing of the same algorithm requires porting that kernel, not the JAX CPU fallback with the CUDA path gated out.

3 VALIDATION AND PERFORMANCE

All timings are single-tenant on one NVIDIA RTX PRO 6000 Blackwell. The CPU reference is NaMaster 3.0 / **ducc0** 0.39.1 on the same host, a 96-core Threadripper PRO 9995WX.

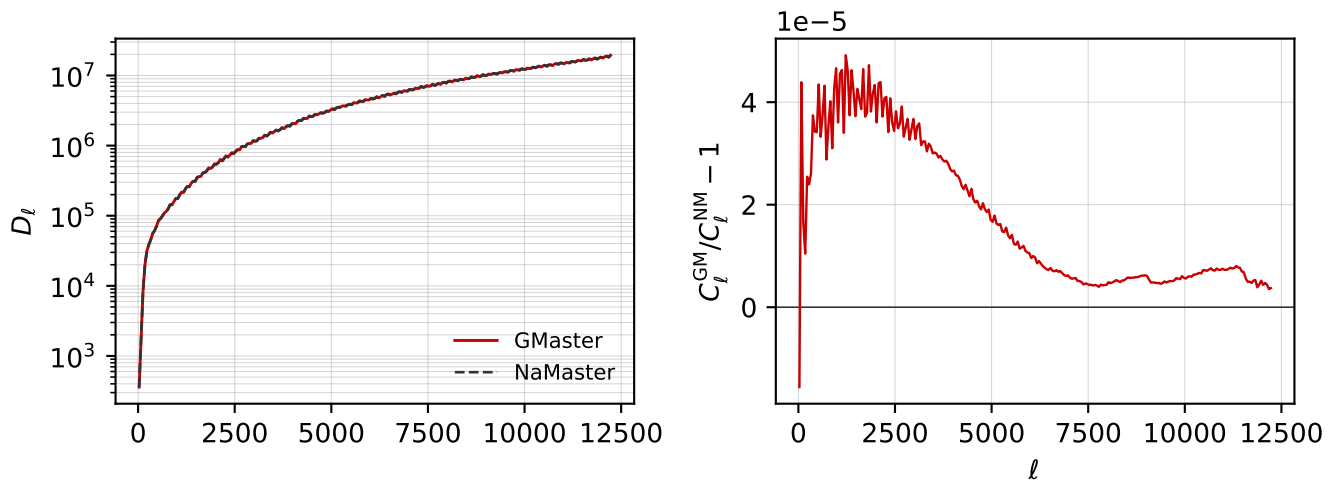


Figure 1. ACT DR6 night PA4 f_{150} source-free coadd (Naess et al. 2025) at $N_{\text{side}} = 4096$. Where map panels are shown, (a) is the survey window used as the mask ($f_{\text{sky}} = 0.481$) and (b) Stokes I on that footprint. (c) Decoupled TT D_ℓ after the full MASTER chain: `NmtField` ($n_{\text{iter}} = 3$ mask refinement) $\rightarrow$ coupling matrix $\rightarrow$ coupled cell $\rightarrow$ decoupling with linear bins of width 50; GMaster (red) and NaMaster (grey) share inputs. GMaster executes the latitudinal march of Section 2 inside every spin-0 analysis/synthesis that builds field and mask alms. (d) Fractional bandpower difference $C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1$. This is a methods overlay, not an ACT science product.

End-to-end totals are warmed medians of field construction, coupling matrix, coupled cell and decoupling. In Table 2, **rel** is the maximum relative deviation of the *coupled* pseudo-spectrum from `pymaster` on the same inputs ($\sim 10^{-7}$ – 10^{-5} in the table). Decoupled bandpower agreement $|C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1|$ on the ACT DR6 scaling board (linear bins of 30, `benchmarks/scaling/results.json`) reaches 7×10^{-3} only in the highest bandpowers at $N_{\text{side}} = 8192$, and stays below 10^{-3} for $\ell < 2N_{\text{side}}$ there (Figure 2, right). Isolated transforms are timed against the `ducc0` call NaMaster issues.

For the scaling board only, ACT maps cached at a given N_{side} are `ud_graded` to lower resolutions when a native file is absent; the overlays in Figure 1 use maps prepared at $N_{\text{side}} = 4096$ and are not downgraded.

Table 2 is the MASTER board on this host. The shipped route is 10.2 – $21.2 \times$ NaMaster at $N_{\text{side}} = 128$ – 2048 in both spins, 5.0 – $5.8 \times$ at $N_{\text{side}} = 64$ (launch-latency bound) and $9.6 \times$ at $N_{\text{side}} = 4096$ spin 0. Stock NaMaster segfaults at $N_{\text{side}} = 4096$ spin 2 on 32-bit coupling-matrix indices; GMaster finishes that cell in 8.8s. One isolated latitudinal pass against `ducc0` is 5.6 – $8.5 \times$ at $N_{\text{side}} = 1024$ – 4096 , with coefficient agreement $\sim 1.2 \times 10^{-7}$. With the march disabled and rings and coupling forced to float64, decoupled bandpowers reproduce 4.8×10^{-13} at $N_{\text{side}} = 256$ spin 0.

Figure 1 is the same estimator on the ACT DR6 night PA4 f_{150} source-free coadd (Naess et al. 2025), Stokes I , analysed at $N_{\text{side}} = 4096$ (maps prepared at that resolution from the public DR6 release). The mask is the survey window (finite non-zero pixels; $f_{\text{sky}} = 0.481$). GMaster and NaMaster share the mask, linear bins of width 50, and $n_{\text{iter}} = 3$. Decoupled TT agrees to $\text{rms}(C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1) = 1.41 \times 10^{-5}$. Wall-clock after compilation: 7.63s versus 70.4s on 192 cores. The FLAMINGO L2p8 unlensed Compton- y lightcone (Schaye et al. 2023), full sky at $N_{\text{side}} = 4096$, gives $\text{rms} 2.32 \times 10^{-7}$ in 8.64s versus 69.5s. Both overlays time only the estimator. Figure 2 contrasts the float32 march floors of Table 1 with

Table 2. End-to-end MASTER on the 96-core host. NM and GM are NaMaster and GMaster milliseconds. **rel** is the maximum relative deviation of the *coupled* spectrum from `pymaster`, not the decoupled bandpower ratio in Figure 2.

N_{side}	spin	NM	GM	Ratio	rel
64	0	9	2	5.8	1.15×10^{-7}
64	2	12	2	5.0	2.72×10^{-7}
128	0	20	1	13.3	4.92×10^{-7}
128	2	40	3	13.1	4.37×10^{-7}
256	0	58	4	15.0	6.36×10^{-7}
256	2	161	8	20.7	6.30×10^{-7}
512	0	314	17	18.1	7.07×10^{-7}
512	2	641	30	21.2	2.99×10^{-6}
1024	0	1682	102	16.5	1.45×10^{-6}
1024	2	3017	212	14.2	7.79×10^{-6}
2048	0	10311	1014	10.2	7.25×10^{-6}
2048	2	17767	1244	14.3	1.19×10^{-5}
4096	0	71583	7458	9.6	7.0×10^{-6}
4096	2	—	8805	—	finite

decoupled bandpower agreement on the ACT scaling board; Figure 3 is the wall-clock board of Table 2.

4 CONCLUSIONS

A three-term Jacobi recurrence in float32 is limited by $1/\sin \phi$ amplification, not by the width of its words. Marching ($v, v - v_{\text{prev}}$) moves the injection of every rounding onto a difference that already vanishes at the turning points, and a two-word C removes the residual coherent drift. Carried as a CUDA kernel behind NaMaster's API, that rewrite is the default latitudinal operator in GMaster and is 10 – $21 \times$ NaMaster at $N_{\text{side}} = 128$ – 2048 on the board of Table 2. The same pipeline on an ACT DR6 coadd and a FLAMINGO Compton- y map at $N_{\text{side}} = 4096$ agrees at 10^{-5} – 10^{-7} in under nine seconds.

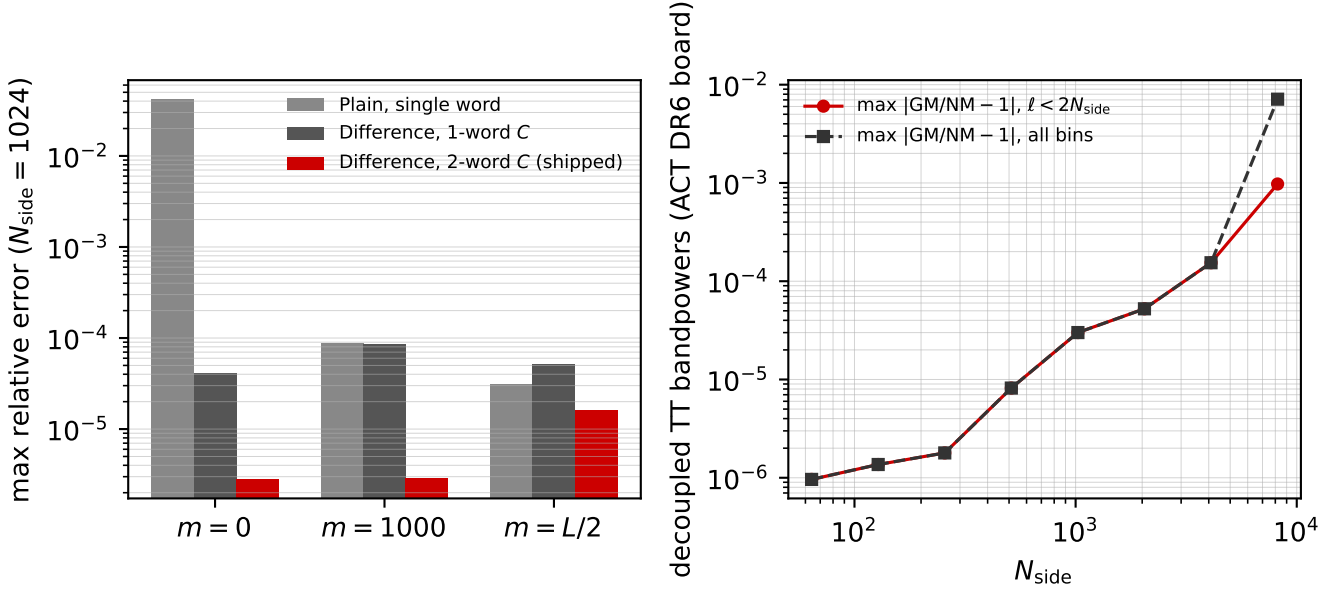


Figure 2. Left: relative error of a float32 march against the float64 recurrence at $N_{\text{side}} = 1024$ spin 0 (Table 1). Right: maximum $|C_{\ell}^{\text{GM}}/C_{\ell}^{\text{NM}} - 1|$ on decoupled TT bandpowers from the ACT DR6 end-to-end scaling board (`results.json`); solid red: bins with $\ell < 2N_{\text{side}}$; dashed grey: all bins. GMaster curves are red; reference curves are black/grey.

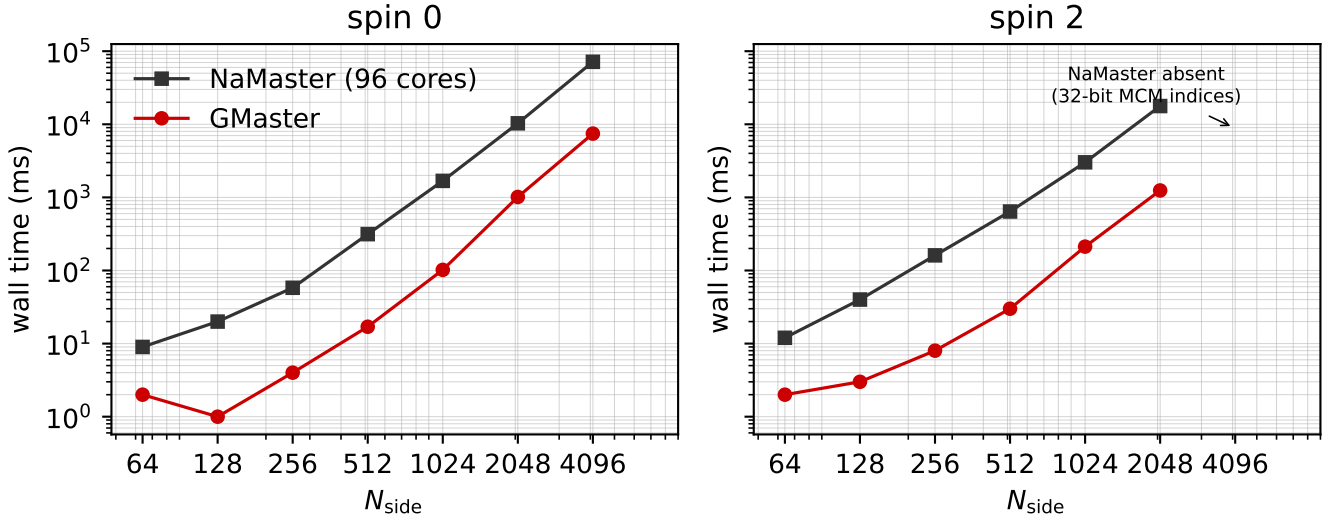


Figure 3. End-to-end MASTER wall time on the host of Table 2. Grey is NaMaster on 96 cores; red is GMaster. NaMaster is absent at $N_{\text{side}} = 4096$ spin 2, where stock NaMaster segfaults on 32-bit coupling-matrix indices.

No other GPU spherical-harmonic implementation was timed on the full MASTER board. The shipped default trades NaMaster's float64 transform contract for decoupled bandpowers that match NaMaster to better than 10^{-3} below $\ell = 2N_{\text{side}}$ at $N_{\text{side}} = 8192$ and up to 7×10^{-3} in the highest bandpowers (Figure 2); the exact float64 route is one flag. The overlays are methods comparisons, not science products.

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NaMaster, DUCC/libsharp, s2fft, threej_cosmo and SHTns are the reference stack against which the numbers were measured. The Lean 4 development uses Mathlib. ACT DR6 maps are those of Naess et al. (2025). Every numerical claim was measured on the hardware of Section 3.

DATA AVAILABILITY

GMaster, the overlay scripts, the committed bandpowers of both overlays and a notebook that rebuilds Figure 1 and the timing table from them are at <https://github.com/licongxu/GMaster>. Figure 1 uses the public ACT DR6 maps (Naess et al. 2025). The FLAMINGO L2p8 Compton- y light-cone is described by Schaye et al. (2023).

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